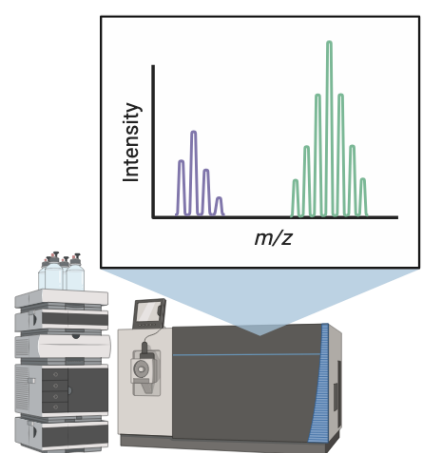
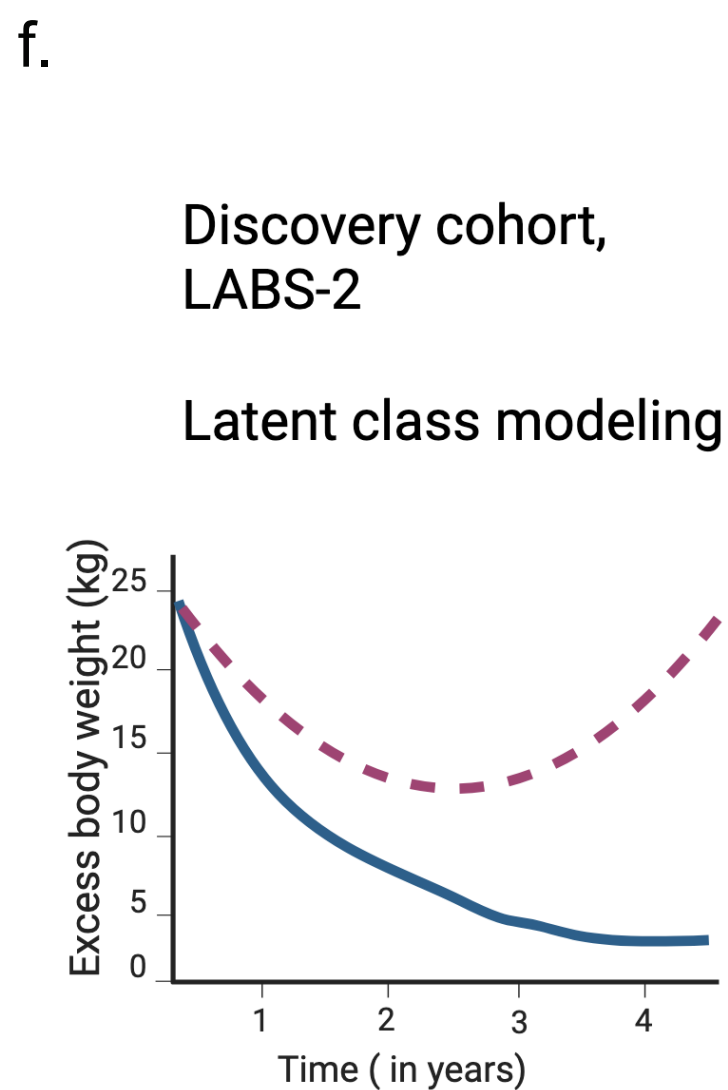
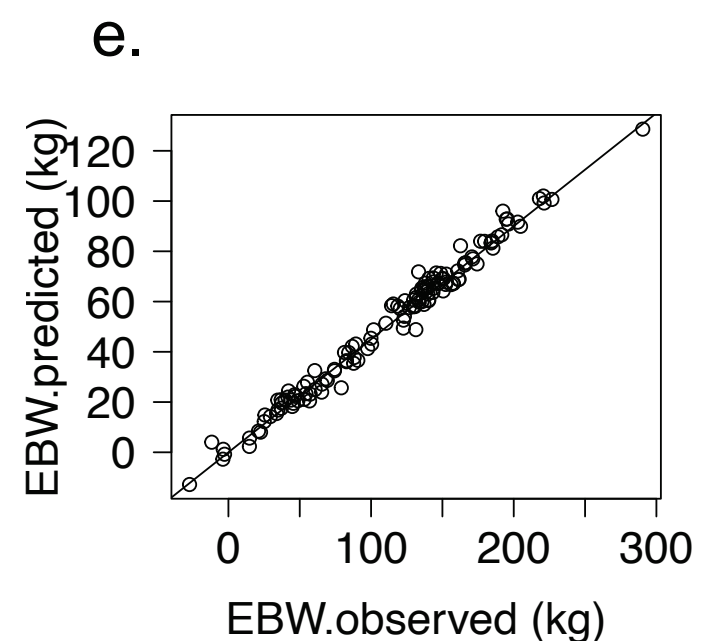
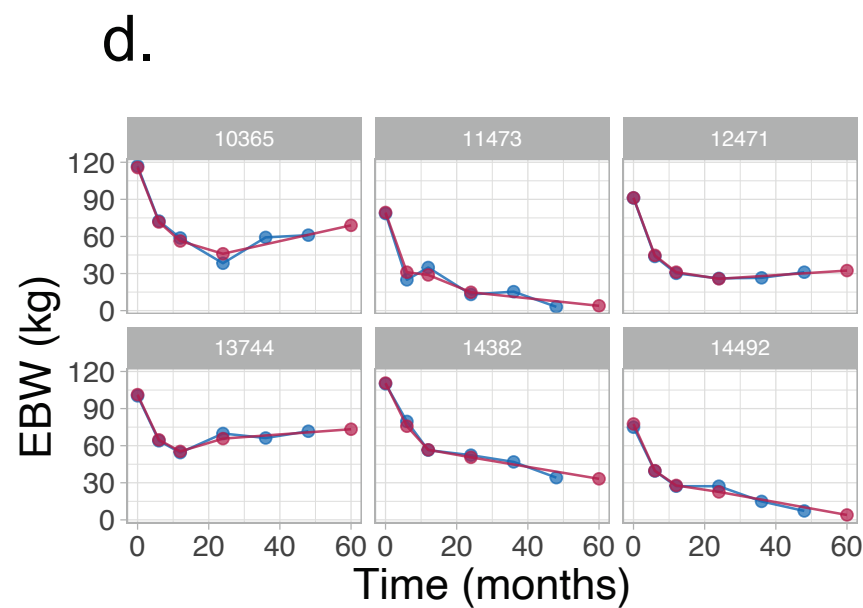
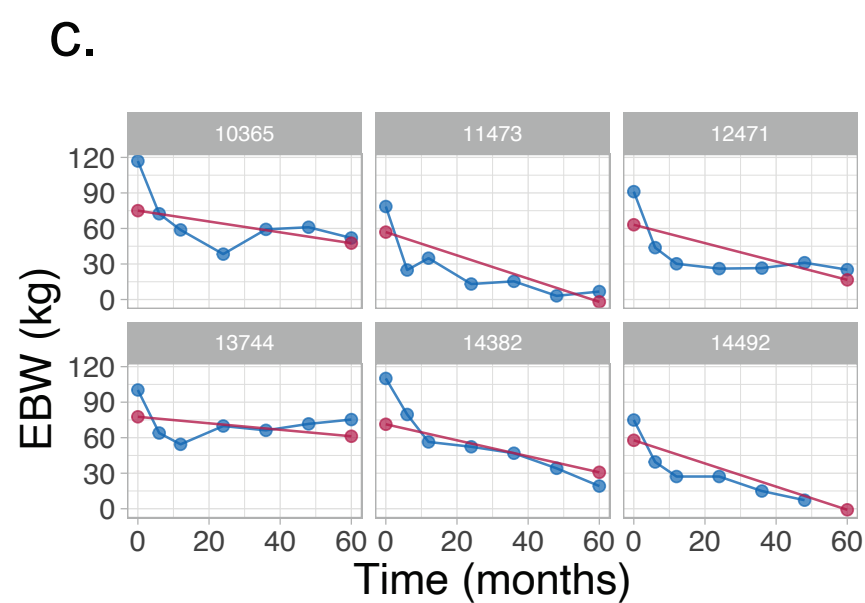
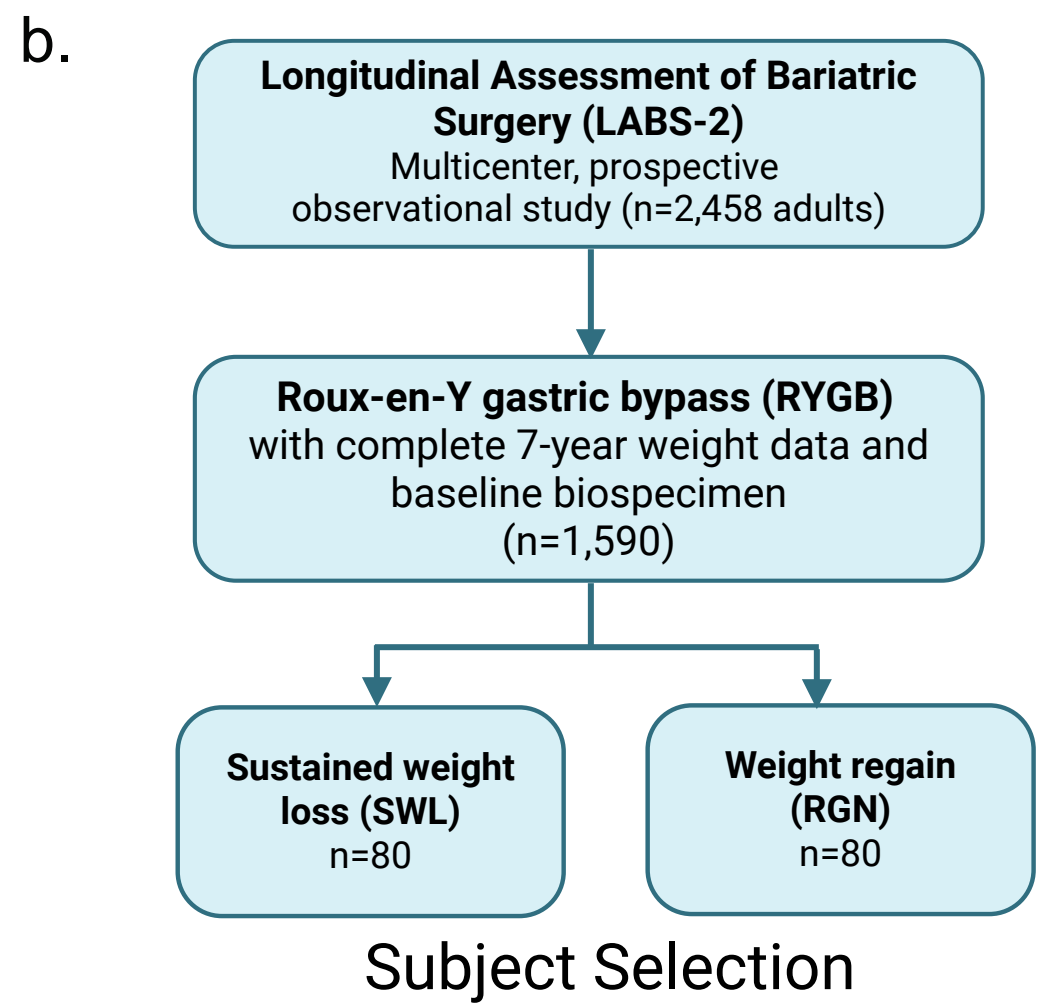
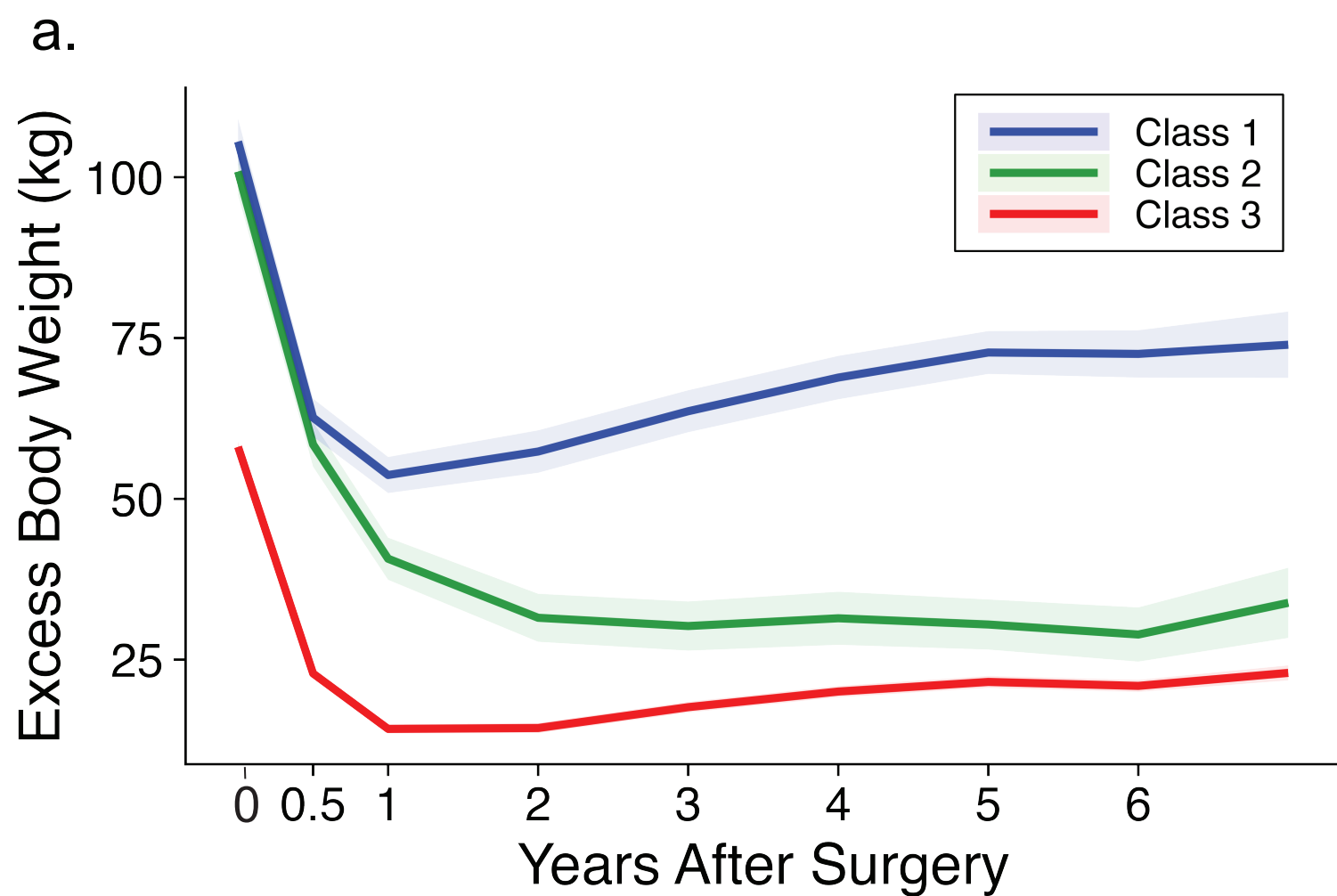
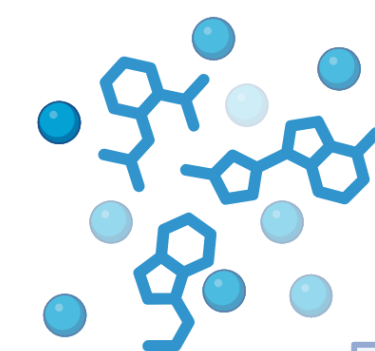




Lipids
Amino acids
Carbohydrates
Xenobiotics
Nucleotide derivatives
Other metabolites



Metabolite risk score (MetRS)

Independent cohorts

Post-RYGB cohort
(n=36, median 7.2 years)

Obesity extremes
(n=198)

Genetic analyses

Framingham Heart Study

Study Design